Question 1

Attach a USB drive to the system and then safely detach it using Linux commands.

Approach

First identify the USB device path, mount it to a directory, then unmount it after use.

Commands

- `lsblk` → shows connected storage devices
- `mount` → attach device to directory
- `unmount` → safely detach device

Output

STEP 1 : identify usb device

```
┌──(aman㉿kali)-[~]
└─$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 30.7G  0 disk
├─sda1   8:1    0 29.1G  0 part /
├─sda2   8:2    0    1K  0 part
└─sda5   8:5    0  1.6G  0 part [SWAP]
sdb      8:16   1 29.3G  0 disk
└─sdb1   8:17   1 29.2G  0 part
sr0     11:0    1 1024M  0 rom
```

STEP 2 : mount usb drive

```
┌──(aman㉿kali)-[~]
└─$ sudo mkdir -p /mnt/usb

┌──(aman㉿kali)-[~]
└─$ sudo mount /dev//sdb1 /mnt/usb
```

STEP 3 : unmount usb drive

```
┌──(aman㉿kali)-[~]
└─$ sudo umount /mnt/usb
```

Key Observations

Linux treats USB drives as block devices like /dev/sbd.
Unmounting prevents data corruption.

What I learned

Devices must be mounted before accessing their files.

Question 2

Find how much storage space is available on the main disk.

Approach

Use a command that displays disk usage and free space in readable format.

Commands

- `df` → shows disk free space
- `df -h` → human readable format

Output

STEP 1 : check disk free space

```
┌──(aman㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            3.8G     0  3.8G   0% /dev
tmpfs           795M 1000K  794M   1% /run
/dev/sda1        29G   17G   11G  62% /
tmpfs           3.9G  4.0K  3.9G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.9G  8.0K  3.9G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           795M  108K  795M   1% /run/user/1000

┌──(aman㉿kali)-[~]
└─$ sudo mount /dev/sdb1 /mnt/usb

┌──(aman㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            3.8G     0  3.8G   0% /dev
tmpfs           795M 1000K  794M   1% /run
/dev/sda1        29G   17G   11G  62% /
tmpfs           3.9G  4.0K  3.9G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.9G  8.0K  3.9G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           795M  108K  795M   1% /run/user/1000
/dev/sdb1        30G   21G  8.6G  71% /mnt/usb
```

Key Observations

Primary disk usually mounted on root(/).
-h flag shows size in GB/MB.

What I learned

Disk usage monitoring helps prevent storage exhaustion.

Question 3

Clone full data from one USB device to another including hidden or deleted content.

Approach

Use low-level copy command to duplicate raw disk blocks.

Commands

- `dd` → copies raw data bit-by-bit

- `lsblk` → identify device names

Output

STEP 1 : identify devices

```
┌──(aman㉿kali)-[~]
└─$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 30.7G  0 disk
├─sda1   8:1    0 29.1G  0 part /
├─sda2   8:2    0    1K  0 part
└─sda5   8:5    0  1.6G  0 part [SWAP]
sr0     11:0    1 1024M  0 rom
```

STEP 2 : copy disk content

```
┌──(aman㉿kali)-[~]
└─$ sudo dd if=/dev/sda1 of=/dev/sdc bs=4M status=progress
[sudo] password for aman:
4005560320 bytes (4.0 GB, 3.7 GiB) copied, 47 s, 85.1 MB/s
dd: error writing '/dev/sdc': No space left on device
968+0 records in
967+0 records out
4057698304 bytes (4.1 GB, 3.8 GiB) copied, 47.5204 s, 85.4 MB/s
```

Key Observations

dd copies raw blocks, not files.
Deleted data may still exist in disk sectors.

What I learned

Bit-level cloning is useful in digital forensics.

Question 4

Display structure and details of storage devices connected to the system.

Approach

Use a command that lists block devices and partitions.

Commands

- `lsblk` → lists block devices

- `lsblk -f` → shows filesystem type

Output

STEP 1 : view block devices

```
┌──(aman㉿kali)-[~]
└─$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 30.7G  0 disk
├─sda1   8:1    0 29.1G  0 part /
├─sda2   8:2    0    1K  0 part
└─sda5   8:5    0  1.6G  0 part [SWAP]
sr0     11:0    1 1024M  0 rom
```

STEP 2 : view filesystem details

```
┌──(aman㉿kali)-[~]
└─$ lsblk -f
NAME   FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1 ext4   1.0         6fbf4013-33f5-4223-830a-92696c84629c   10.4G    58% /
├─sda2
└─sda5 swap   1           8f40bb2d-2ea7-4208-a5d9-3b2707cd771e               [SWAP]
sr0
```

Key Observations

Block devices include disks, USB, partitions.
Shows mount point and filesystem type.

What I learned

Understanding storage layout helps in system administration.

Security Insights

1. Improper mounting/unmounting can expose data corruption risks.
2. Monitoring disk usage prevents denial-of-service via storage exhaustion.
3. Raw disk cloning allows forensic recovery of deleted evidence.
4. Block device enumeration reveals system storage attack surface.